Mentoring Mobile App (Morocco) — Project Plan

Project Overview and Goals

- Core objectives and scope
  - Connect experienced mentors and mentees across Morocco with a Tinder-like swipe interface for mentor discovery and chat-enabled communication.
  - Promote mentor visibility via profiles and a feed for short advice posts, all in a modern 100% dark mode UI.
  - Provide intelligent matching via rules-based logic and optional ML-based recommendations using textual embeddings and similarity scores.
- Target users and value proposition
  - Mentors: showcase photo, name, sector, skills, years experience, short bio, languages, location to attract mentees.
  - Mentees: create profiles with desired sector, skills, objectives to discover compatible mentors and receive personalized recommendations.
  - Value: fast discovery, secure chat, compatibility scores, and mentor-sourced micro-content in a dark-mode UX.
- Success metrics and test dataset
  - Metrics: match rate, message initiation rate, retention (7/30-day), recommendation precision (offline).
  - Test dataset: initial pilot with ~50 mentors and ~50 mentees to validate matching rules and ML pipelines.

Functional Specifications and UI

- Authentication and account flows
  - Sign-up as mentee or mentor with Firebase Auth supporting email/password and secure session management.
  - Password reset, login/logout flows, and account editing with role-specific profile fields.
  - Audit: secure token handling, session expiration, and account deletion option.
- Profile schemas and edit UX
  - Mentor fields: photo, name, sector, skills, years experience, short bio, languages, location; editable anytime.
  - Mentee fields: photo, name, desired sector, skills, objectives, languages, location; editable anytime.
  - Profile detail view: full mentor profile accessible from swipe cards with contact/match actions.
- Main feed and dark-mode UI
  - Tinder-style swipe deck showing mentor cards with photo, name, sector, key skills, years, short bio, compatibility score.
  - Swipe left = reject, swipe right = interested; persistent dark-mode styling across all views.
  - Mentor posts feed: short posts or tips shown in timeline with like/comment affordances.

Matching, Recommendations, and ML

- Rules-based matching and compatibility score
  - Basic rules: sector match + overlapping skills weightings produce a 0–100 compatibility score per mentee-mentor pair.
  - Output: ranked recommended mentors list per mentee and displayed score on swipe cards.
  - Tuning: adjustable weights, minimum thresholds, and business rules for region/language constraints.
- Advanced ML recommendations and embeddings
  - Pipeline: text preprocessing of bios/skills → transformer embeddings (e.g., sentence-transformers) → FAISS index for nearest neighbors.
  - Similarity: cosine similarity on embeddings to augment rules-based score and produce hybrid ranking.
  - Tools: Python, pandas, scikit-learn, transformers, FAISS; offline evaluation on pilot data.
- Predictive scoring and evaluation
  - Option: train a predictive compatibility model using labeled interactions (likes, messages) to estimate match probability.
  - Metrics: precision@k, recall, AUC; offline validation with cross-validation and pilot logs.
  - Monitoring: periodic retraining and drift checks, incremental indexing for new profiles.

Messaging, Notifications, and Data

- Real-time chat and message flows
  - Chat implemented via Firebase Firestore with real-time listeners, message threads between mentor and mentee.
  - UX: dark-mode chat UI, read receipts, timestamps, and message pagination for performance.
  - Security: Firestore rules to restrict access to involved participants only.
- Push notifications and alerts
  - Push via Firebase Cloud Messaging for new messages and new matches; notifications formatted for dark-mode displays.
  - In-app alerts for matches, recommendations updates, and mentor posts; user opt-in management available.
  - Throttling: rate limits and batching to avoid notification fatigue.
- Data collection, privacy, and test dataset
  - Store profiles, interactions, messages (with retention policies) in Firestore; anonymize logs for ML training.
  - Pilot dataset: seed 50 mentors / 50 mentees, collect interactions for model training and evaluation.
  - Privacy: GDPR-like best practices, consent for data use, and option to export/delete user data.

Technical Architecture and Implementation Plan

- Tech stack and integrations
  - Frontend: Flutter for iOS & Android with unified dark-mode component library.
  - Backend: Firebase Firestore for data, Firebase Auth for authentication, FCM for notifications.
  - ML: Python services for embedding generation and FAISS indexing; models served offline or via lightweight API.
- Development milestones and timeline
  - Phase 1 (4 weeks): core auth, profile CRUD, swipe UI, Firestore schema, dark-mode components.
  - Phase 2 (4 weeks): real-time chat, notifications, mentor posts feed, basic rules-based matching.
  - Phase 3 (4–6 weeks): ML pipeline, embeddings & FAISS integration, compatibility scoring, evaluation on pilot dataset.
- Deployment, monitoring, and maintenance
  - CI/CD: automated builds for Flutter, Firebase hosting for functions (if any), and scripted ML pipeline runs.
  - Monitoring: analytics for matches/messages, error reporting, ML model performance dashboards.
  - Maintenance: scheduled retraining, dataset growth plan, and UI/UX adjustments based on user feedback.